ARCHIT AGRAWAL

San Francisco Bay Area | 623-312-0435 | architagrawal000@gmail.com | LinkedIn | GitHub | Portfolio

PROFESSIONAL EXPERIENCE

EdPlus, Arizona State University **06/2026 - Present**
AI Software Engineer *Phoenix, AZ*

- Own end-to-end architecture and delivery of an AWS analytics platform for any structured dataset: a 14-page Nuxt frontend, 70 NestJS API routes and shared TypeScript contracts.
- Cut end-to-end AWS pipeline runtime from 500 to 85 seconds by removing managed-agent startup overhead and invoking the underlying tools directly.
- Accelerated publishing of a 51,280-respondent survey from 19 minutes to 109 seconds; an independent raw-data audit matched all evaluated figures.
- Engineered 64-way AWS Step Functions processing with a failure circuit breaker after diagnosing silent row loss in runs that reported success.
- Delivered interactive chart sorting, filtering, top-N and undo in Nuxt; in-process DuckDB kept measured query latency flat as input rows grew 200x.
- Reached 100% join integrity on repeat runs with schema-constrained LLM output and per-row IDs; the prior pipeline silently mislabeled 1 in 8 rows.
- Secured the analytics proof of concept with entitlement-gated row-level access, Bedrock Guardrails deployed through AWS CDK and enforced per-stage spend limits.

MyStage Music Inc. **03/2025 - 06/2026**
Founding AI/ML Engineer *Remote*

- Delivered sub-50 ms venue search through FastAPI and Algolia, synchronizing Firestore updates into the search index in real time.
- Expanded Playwright ingestion to 70,000+ records daily from 650+ venues, using proxy rotation, backoff and transactional locking to handle unreliable sources.
- Reduced production service dependencies 64% by consolidating a 14-Cloud-Function backend into a checkpointed Python/LangGraph workflow on GCP Cloud Run.
- Implemented pause-and-resume for incomplete records and retry-safe task processing through deterministic IDs and atomic Firestore claims.
- Covered routing, transactional claims, retries and checkpoint resumes with 200+ pytest unit and integration tests; traced workflow execution through Logfire.

EdPlus, Arizona State University **09/2023 - 05/2025**
Software Engineer *Phoenix, AZ*

- Built a multi-tenant course assistant from React UI to retrieval services, used by 1,000+ faculty creating courses for 60,000+ students.
- Increased assessment-tool engagement 35% and reduced bounce 20% by building React and Material UI workflows for bulk import, question reuse and revision comparison.
- Enabled configuration-based college onboarding through tenant-scoped retrieval, keeping each college's documents within its own search boundary.
- Cut transcript-analysis time from four hours to 15 minutes with a Neo4j knowledge graph and schema-validated, LLM-generated Cypher queries.
- Established Prompt Flow release checks for groundedness, relevance and coherence, evaluating prompt changes against measured results before deployment.

Knowledge Exchange for Resilience, Arizona State University **06/2024 - 08/2024**
Data Research Aide *Phoenix, AZ*

- Delivered 20+ production REST APIs using .NET 8, Dapper and MediatR, with JWT authentication and AWS SQS for queued work.
- Cut API latency from 198 to 20 ms with Redis caching and SQL rewrites, verified under simulated peak load with Locust.
- Built an idempotent ETL pipeline for 10,000+ researcher profiles daily, using PostgreSQL natural-key upserts to prevent duplicates after partial failures.

- Enforced uniqueness, not-null and range constraints in PostgreSQL to reject invalid records during ingestion.

Zeus Learning **01/2022 - 07/2023**
Software Engineer *Mumbai, India*

- Migrated a C#/.NET monolith to Dockerized Kubernetes microservices, reducing infrastructure cost 20% and resource footprint 35%.
- Reduced deployment time 70% and production incidents 40% through containerized CI/CD and SonarQube quality gates.
- Built a Redis-backed desk-booking service used across 300+ offices at Fortune 500 companies.
- Improved student-listing performance 30% with pagination and S3 asset delivery; reduced class-detail latency 10% through SQL query rewrites.

EAT.FIT **09/2021 - 12/2021**
Product Intern *Bengaluru, India*

- Cut driver-tracking server cost 45% with distance-based polling and client-side interpolation that kept map movement smooth.
- Built a WebSocket and Express notification service with backpressure-aware fan-out and duplicate-safe redelivery; reported CSAT rose 28% and support volume fell 35%.

PROJECTS

PrismSplit **2025 - Present**

- Built a TypeScript/React Native expense app with transactional PostgreSQL billing and settlement, integer-cent accounting and offline write persistence.
- Added idempotency keys, real-time Supabase subscriptions and versioned persistence to handle retries and cached data across app upgrades.
- Wrote Jest unit and interaction tests, Maestro end-to-end flows and Android release-build checks.

No-Code Pipeline Builder **2025**

- Built a React/ReactFlow workflow editor with typed nodes, undo/redo and autosave, backed by FastAPI graph validation.
- Tested editor history, persistence and keyboard interactions with Jest, React Testing Library and property-based tests.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C#
Frontend: React, Next.js, Vue/Nuxt, React Native, ReactFlow, Zustand, Material UI
Backend & APIs: Node.js, NestJS, FastAPI, Express, .NET 8, REST APIs, WebSockets, JWT
Databases & Search: PostgreSQL, Redis, DuckDB, Firestore, Algolia, Supabase, Neo4j, pgvector
Cloud & DevOps: AWS (Lambda, S3, SQS, Step Functions, CDK), GCP Cloud Run, Docker, Kubernetes, CI/CD
Testing & Observability: pytest, Jest, Vitest, React Testing Library, Playwright, Maestro, Locust, SonarQube, Logfire
AI Applications: AWS Bedrock, LangGraph, RAG, LLM evaluation, MCP

EDUCATION

Arizona State University (ASU) **08/2023 - 05/2025**
M.S., Computer Science | GPA: 4.0 *Tempe, AZ*

Dhirubhai Ambani Institute of Information and Communication Technology **08/2018 - 05/2022**
B.Tech., Information and Communication Technology *Gandhinagar, India*